

## Class XI | Unit 7 | weight 15%

# Computer communication and networks

Board weight about 15% | 25 theory periods

A network is computers and other devices joined so they can exchange data. This lecture names the parts of a communication, the directions data can travel, the media, the shapes of a network, and the layered models used to describe it.

## What you should be able to do

- Describe a network, its advantages, and the five communication components.
- Define simplex, half duplex and full duplex.
- Compare guided and unguided media.
- Describe LAN and WAN, and the bus, star, ring, mesh, tree and hybrid topologies.
- Name ISO, IEEE, ITU and ASCII, and describe the OSI, TCP/IP and Internet models.

## Path through the lecture

- 1 Start from a message the class actually sends, and name sender, receiver, medium, message and protocol on that example.
- 2 Teach direction with a television, a walkie-talkie and a telephone.
- 3 Draw bus, star and ring. Describe mesh, tree and hybrid from the drawings.
- 4 Build the OSI stack from the physical wire up to the program the user touches. Place TCP/IP beside it.

## 7.1 A network and its parts

**Computer network.** A collection of computers and other devices connected so that they can communicate and share resources.

A college network shares printers, shares files, lets staff send messages, keeps a backup in one place, and gives many computers a path to the internet. The sharing has to be organised, and the cable or wireless link has to be protected and maintained.

### Five components of communication

| Component | In a college email                                               |
|-----------|------------------------------------------------------------------|
| Sender    | The computer of the person who writes the message                |
| Receiver  | The computer of the person who reads it                          |
| Medium    | The cable or the wireless link that carries the signal           |
| Message   | The mail itself                                                  |
| Protocol  | The agreed rules, such as the rules that format and deliver mail |

**Protocol.** A set of rules that communicating devices follow so that a message sent by one can be understood by the other.

## 7.2 Direction of data

### Communication modes

| Mode    | Direction          | Example                                                       |
|---------|--------------------|---------------------------------------------------------------|
| Simplex | One direction only | A keyboard sending to the computer, or a television broadcast |

| Mode        | Direction                        | Example                                         |
|-------------|----------------------------------|-------------------------------------------------|
| Half duplex | Both directions, one at a time   | A walkie-talkie                                 |
| Full duplex | Both directions at the same time | A telephone conversation, and ordinary Ethernet |

### 7.3 Transmission media

**Guided medium.** A physical path that contains the signal: a wire or a fibre.

**Unguided medium.** A wireless path. The signal travels through the air or space.

- **Twisted pair** is the ordinary copper LAN cable. Wires are twisted to reduce interference. Shielded twisted pair adds a foil or braid. Unshielded twisted pair is the common Ethernet cable.
- **Coaxial cable** has a central copper core, insulation, a shield and a jacket. It resists interference and was widely used for cable television and older networks.
- **Fibre optic** cable carries pulses of light in a glass or plastic core. It carries data over long distances with little electrical interference and with high capacity.
- **Radio waves** travel in many directions and pass through ordinary walls. They carry Wi-Fi and many mobile signals.
- **Microwaves** travel in a straight line between antennas and are used for long point-to-point links. The antennas need a clear path.
- **Infrared** is a short-range light signal, used by some remote controls. It does not pass through walls.

### 7.4 Types and topologies

A **LAN**, local area network, covers a room, a lab or a building and is owned by one organisation. A **MAN**, metropolitan area network, covers a town or a city. A **WAN**, wide area network, covers a large region or the world. The internet is the WAN students use every day. The unit list concentrates on LAN and WAN. MAN is the standard type between them and is in the curriculum's own acronyms.

**Topology.** The arrangement of devices and links in a network.

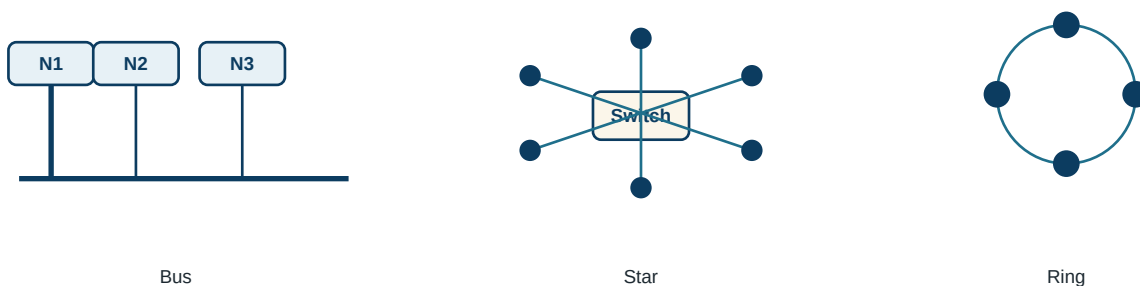


Figure 7.1 Bus, star and ring.

### Topologies

| Shape | How it is joined                                   | What to say in an answer                                         |
|-------|----------------------------------------------------|------------------------------------------------------------------|
| Bus   | Every device taps one backbone cable               | Easy to add a device. A break in the backbone stops the network. |
| Star  | Every device has its own cable to a central switch | A broken spoke affects one device. The centre is essential.      |

| Shape  | How it is joined                             | What to say in an answer                                            |
|--------|----------------------------------------------|---------------------------------------------------------------------|
| Ring   | Each device connects to the next in a circle | A token or a turn controls who may send. A break can stop the ring. |
| Mesh   | Devices have several paths to each other     | The network survives a broken link. It uses more cable.             |
| Tree   | Stars joined in a hierarchy                  | It matches a campus: rooms, floors, then a core switch.             |
| Hybrid | A mixture of the shapes above                | A real college network is often a hybrid, not a pure shape.         |

## 7.5 Standards and architectures

- **ISO**, the International Organization for Standardization, publishes international standards, including the OSI model.
- **IEEE**, the Institute of Electrical and Electronics Engineers, publishes the 802 standards for local networks. IEEE 802.3 is Ethernet. IEEE 802.11 is wireless LAN.
- **ITU**, the International Telecommunication Union, publishes standards for telecommunications.
- **ASCII** is a character-encoding standard, not an organisation. It assigns numbers to letters, digits and controls. The basic set uses 7 bits and contains 128 characters. It is in this unit because communicating computers must agree on how characters are coded.

OSI model



TCP/IP model

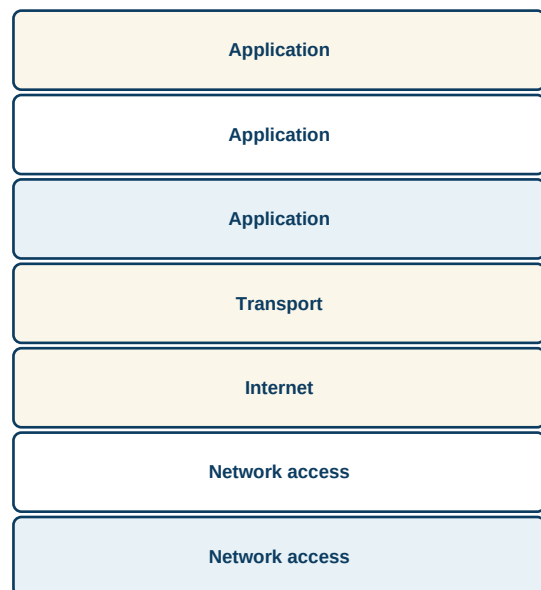


Figure 7.2 OSI beside TCP/IP. Three OSI layers sit inside the TCP/IP application layer.

### The OSI layers, from the wire upward

| Layer          | Job in one sentence                                                                  |
|----------------|--------------------------------------------------------------------------------------|
| 1 Physical     | Sends bits over the medium as electrical, light or radio signals.                    |
| 2 Data link    | Organises bits into frames, uses a hardware address, and detects errors on one link. |
| 3 Network      | Addresses and routes packets across networks. IP works here.                         |
| 4 Transport    | Carries data end to end between programs. TCP and UDP work here.                     |
| 5 Session      | Opens, manages and closes a dialogue between applications.                           |
| 6 Presentation | Translates formats, and can compress or encrypt.                                     |

| Layer         | Job in one sentence                                                         |
|---------------|-----------------------------------------------------------------------------|
| 7 Application | Provides the service the user sees, such as the web, mail or file transfer. |

The **TCP/IP model**, which is the **Internet model** used on the real internet, has four layers. The application layer covers the work of OSI application, presentation and session. The transport layer carries TCP and UDP. The internet layer carries IP and routing. The network-access layer covers the data link and the physical connection. Learn both models. The board may ask for seven OSI names in order, or for the four TCP/IP names.

#### In the lab

Practical work for this unit is in the journal: straight and crossover cables, the devices in a lab network, an IP address with mask and gateway, and the difference between a network with a central server and a network of equal peers.

#### Exam point

Memorise the OSI layers with a fixed sentence you can write under pressure, from layer 7 down to layer 1: All People Seem To Need Data Processing. Then still be able to start from the physical layer if the question says 'from the bottom'.

## Key terms

| Term        | Meaning to write in an answer                               |
|-------------|-------------------------------------------------------------|
| Protocol    | Rules two devices follow in order to communicate.           |
| Simplex     | Communication in one direction only.                        |
| Full duplex | Communication in both directions at the same time.          |
| LAN         | A network covering a small area, owned by one organisation. |
| WAN         | A network covering a large geographical area.               |
| Topology    | The layout of nodes and links.                              |
| OSI model   | A seven-layer reference model for network functions.        |
| TCP/IP      | The four-layer model used by the internet.                  |

## Bring it together

- Communication needs a sender, a receiver, a medium, a message and a protocol.
- Simplex, half duplex and full duplex describe direction.
- Twisted pair, coax and fibre are guided. Radio, microwave and infrared are unguided.
- LAN and WAN describe the area. Bus, star, ring, mesh, tree and hybrid describe the shape.
- OSI has seven layers. TCP/IP, the Internet model, has four.

## Do this in the practical

- 1 Identify straight-through and crossover cables and the devices in the lab: NIC, switch, router, modem.
- 2 Read an IP address, subnet mask and gateway from a working computer.
- 3 Say whether the lab is organised around a central server or as peers.

## Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

**1.** A walkie-talkie is an example of:

- A. Simplex
- B. Half duplex
- C. Full duplex
- D. A topology

**2.** Fibre optic cable carries the signal as:

- A. Pulses of light
- B. Ink
- C. A paper tape
- D. A process state

**3.** In a star topology, the central device is important because:

- A. A failure there affects the whole star
- B. It stores ROM programs
- C. It replaces the protocol
- D. It is a simplex printer

**4.** IEEE 802.11 describes:

- A. Wireless LAN
- B. A C++ data type
- C. EPROM erasing
- D. File attributes

**5.** Which OSI layer routes packets between networks?

- A. Physical
- B. Network
- C. Session
- D. Application

**6.** TCP and UDP belong to the OSI:

- A. Transport layer
- B. Physical layer
- C. A cable colour
- D. CMOS battery

**7.** ASCII is:

- A. A character-encoding standard
- B. A topology
- C. A type of RAM
- D. An operating system

**8.** A LAN is best described as:

- A. A network over a small area under one organisation
- B. A single keyboard
- C. A layer of OSI
- D. A C++ loop

## Answer key

| Q | Answer | Why this is the answer                                                                 |
|---|--------|----------------------------------------------------------------------------------------|
| 1 | B      | Either person can talk, and only one talks at a time.                                  |
| 2 | A      | Fibre carries light pulses in a glass or plastic core.                                 |
| 3 | A      | Every spoke depends on the centre.                                                     |
| 4 | A      | IEEE 802.11 is the wireless LAN family of standards.                                   |
| 5 | B      | The network layer handles logical addressing and routing.                              |
| 6 | A      | Transport provides end-to-end communication. TCP and UDP are its well-known protocols. |
| 7 | A      | ASCII assigns numeric codes to characters.                                             |
| 8 | A      | LAN means local area network.                                                          |

## Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

**Q1.** Name the five components of data communication.

### Model answer

Sender, receiver, medium, message and protocol. The sender creates the message, the medium carries it, the receiver takes it, and the protocol is the set of rules that makes the exchange understandable.

**Q2.** Differentiate half duplex and full duplex.

### Model answer

Half duplex allows both directions, but only one direction at a time, as in a walkie-talkie. Full duplex allows both directions at the same time, as in a telephone call. Both are two-way. They differ in whether the two directions overlap.

**Q3.** Differentiate guided and unguided media with one example each.

### Model answer

A guided medium contains the signal in a physical path, for example fibre optic cable. An unguided medium carries the signal through the air, for example radio waves used by Wi-Fi. Guided media follow a cable. Unguided media need no cable between the devices.

**Q4.** Give one advantage and one risk of bus topology.

### Model answer

Cable use is simple: devices share one backbone, and a new device can be tapped onto it. A break in that backbone interrupts the network for every device on it.

**Q5.** Write the four layers of the TCP/IP model.

### Model answer

Application, transport, internet and network access. The application layer covers the OSI application, presentation and session work. Transport carries TCP and UDP. The internet layer routes IP packets. Network access puts frames and bits onto the local link.

## Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

**Question 1.** Explain types of networks and network topologies with a labelled description of each topology.

- 1 Define a network and distinguish LAN, MAN and WAN by area.
- 2 Define topology.
- 3 Describe bus, star and ring, each with how devices are joined and one strength and one weakness.
- 4 Describe mesh, tree and hybrid in the same way.
- 5 Say which shape matches a school lab and why. A star, or a tree of stars, is the usual answer.

**Question 2.** Explain the OSI model and compare it with TCP/IP.

- 1 Say that OSI is a reference model of seven layers and name them in order.
- 2 Give one job for each layer, especially physical, data link, network, transport and application.
- 3 Name the four TCP/IP layers and show which OSI layers they cover.
- 4 State that TCP/IP is the model the internet uses. OSI is the teaching and design reference.